

WS N°3: Phase Diagrams (Part II)

1. Is it possible to have a copper–silver alloy of composition 50 wt% Ag–50 wt% Cu that, at equilibrium, consists of α and β phases having mass fractions $W_\alpha = 0.60$ and $W_\beta = 0.40$? If so, what will be the approximate temperature of the alloy ? If such an alloy is not possible, explain why.

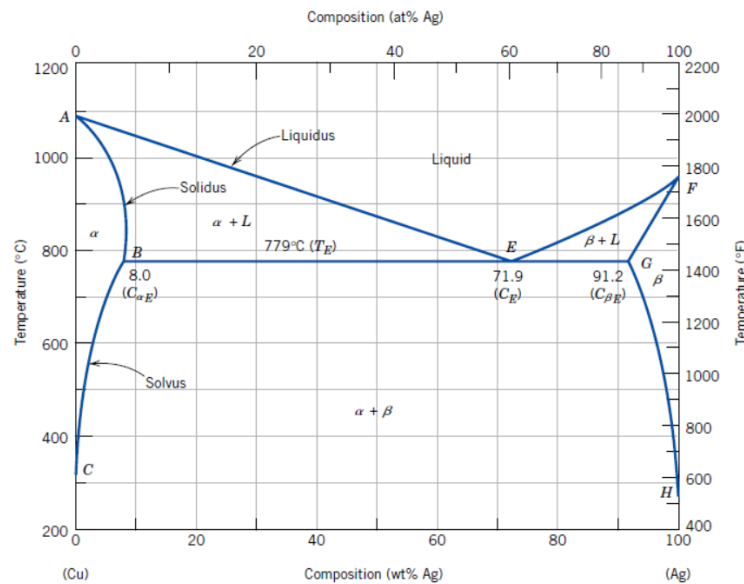


Figure 9.7 The copper–silver phase diagram. [Adapted from *Binary Alloy Phase Diagrams*, 2nd edition, Vol. 1, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]

Solution

It is *not possible* to have a Cu–Ag alloy of composition 50 wt% Ag–50 wt% Cu which consists of mass fractions $W_\alpha = 0.60$ and $W_\beta = 0.40$. Using the appropriate phase diagram, Figure 9.7, and, using Equations 9.1 and 9.2 let us determine W_α and W_β at just below the eutectic temperature and also at room temperature. At just below the eutectic, $C_\alpha = 8.0$ wt% Ag and $C_\beta = 91.2$ wt% Ag; thus,

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{91.2 - 50}{91.2 - 8} = 0.50$$

$$W_\beta = 1.00 - W_\alpha = 1.00 - 0.50 = 0.50$$

Furthermore, at room temperature, $C_\alpha = 0$ wt% Ag and $C_\beta = 100$ wt% Ag; employment of Equations 9.1 and 9.2 yields

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{100 - 50}{100 - 0} = 0.50$$

And, $W_\beta = 0.50$. Thus, the mass fractions of the α and β phases, upon cooling through the $\alpha + \beta$ phase region will remain approximately constant at about 0.5, and will never have values of $W_\alpha = 0.60$ and $W_\beta = 0.40$ as called for in the problem.

2. A 45 wt% Pb–55 wt% Mg alloy is rapidly quenched to room temperature from an elevated temperature in such a way that the high temperature microstructure is preserved. This microstructure is found to consist of the α phase and Mg_2Pb , having respective mass fractions of 0.65 and 0.35. Determine the approximate temperature from which the alloy was quenched.

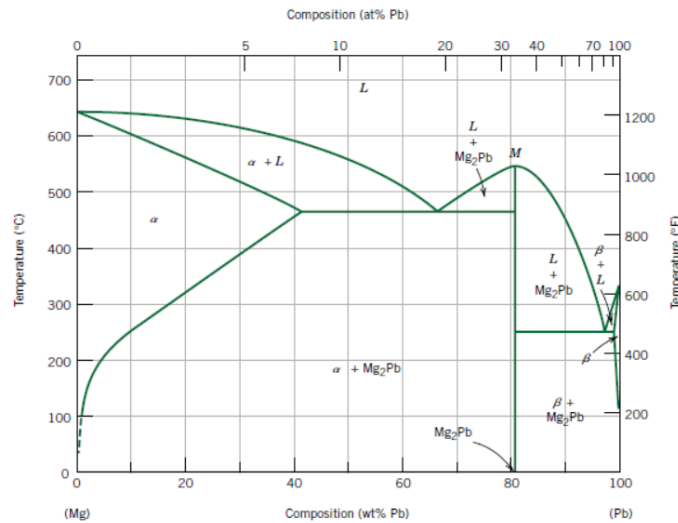


Figure 9.20 The magnesium–lead phase diagram. [Adapted from *Phase Diagrams of Binary Magnesium Alloys*, A. A. Nayeb-Hashemi and J. B. Clark (Editors), 1988. Reprinted by permission of ASM International, Materials Park, OH.]

Solution

We are asked to determine the approximate temperature from which a 45 wt% Pb–55 wt% Mg alloy was quenched, given the mass fractions of α and Mg_2Pb phases. We can write a lever-rule expression for the mass fraction of the α phase as

$$W_{\alpha} = 0.65 = \frac{C_{\text{Mg}_2\text{Pb}} - C_0}{C_{\text{Mg}_2\text{Pb}} - C_{\alpha}}$$

The value of C_0 is stated as 45 wt% Pb–55 wt% Mg, and $C_{\text{Mg}_2\text{Pb}}$ is 81 wt% Pb–19 wt% Mg, which is independent of temperature (Figure 9.20); thus,

$$0.65 = \frac{81 - 45}{81 - C_{\alpha}}$$

which yields

$$C_{\alpha} = 25.6 \text{ wt\% Pb}$$

The temperature at which the α –($\alpha + \text{Mg}_2\text{Pb}$) phase boundary (Figure 9.20) has a value of 25.6 wt% Pb is about 360°C (680°F).

3. For a copper–silver alloy of composition 25 wt% Ag–75 wt% Cu and at 775°C (1425°F), do the following:

- (a) Determine the mass fractions of α and β phases.
- (b) Determine the mass fractions of primary α and eutectic microconstituents.
- (c) Determine the mass fraction of eutectic α .

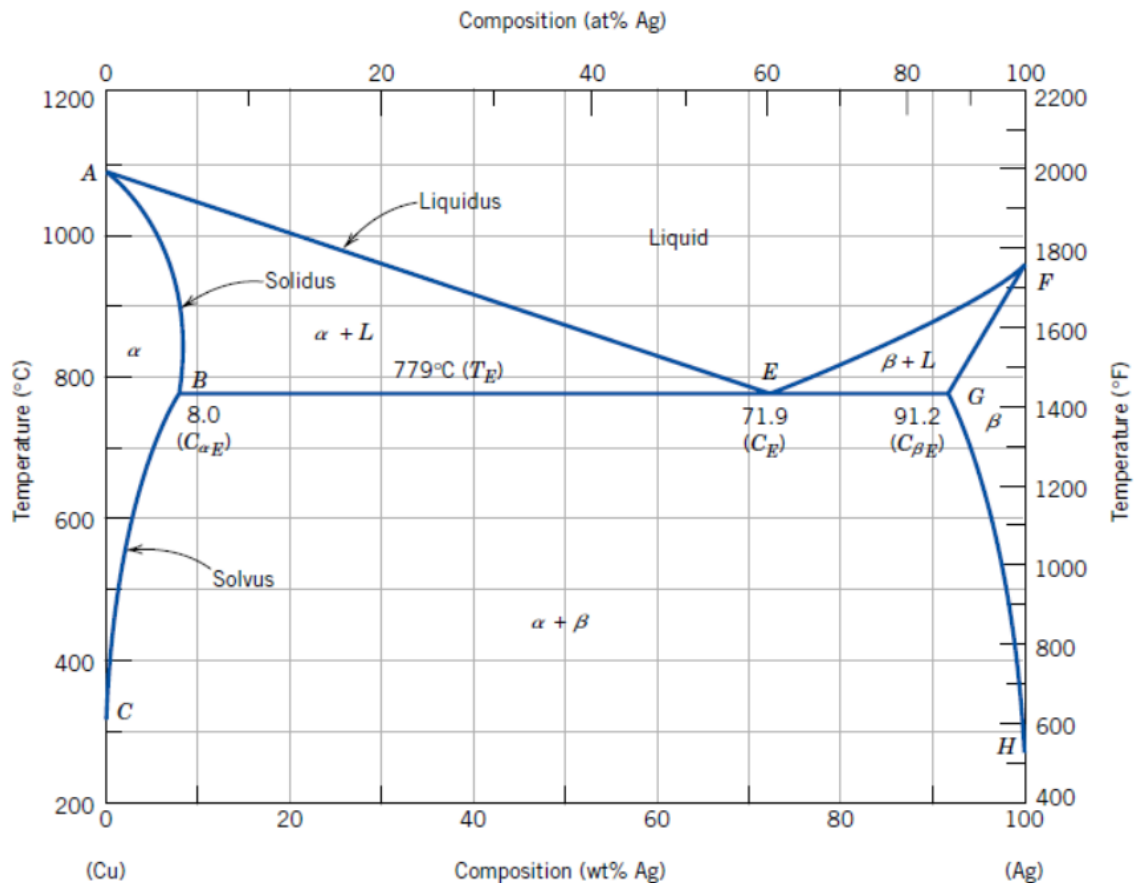


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Solution

(a) This portion of the problem asks that we determine the mass fractions of α and β phases for an 25 wt% Ag-75 wt% Cu alloy (at 775°C). In order to do this it is necessary to employ the lever rule using a tie line that extends entirely across the $\alpha + \beta$ phase field. From Figure 9.7 and at 775°C, $C_\alpha = 8.0$ wt% Ag, $C_\beta = 91.2$ wt% Ag, and $C_{\text{eutectic}} = 71.9$ wt% Sn. Therefore, the two lever-rule expressions are as follows:

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{91.2 - 25}{91.2 - 8.0} = 0.796$$

$$W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{25 - 8.0}{91.2 - 8.0} = 0.204$$

(b) Now it is necessary to determine the mass fractions of primary α and eutectic microconstituents for this same alloy. This requires us to utilize the lever rule and a tie line that extends from the maximum solubility of Ag in the α phase at 775°C (i.e., 8.0 wt% Ag) to the eutectic composition (71.9 wt% Ag). Thus

$$W_{\alpha'} = \frac{C_{\text{eutectic}} - C_0}{C_{\text{eutectic}} - C_\alpha} = \frac{71.9 - 25}{71.9 - 8.0} = 0.734$$

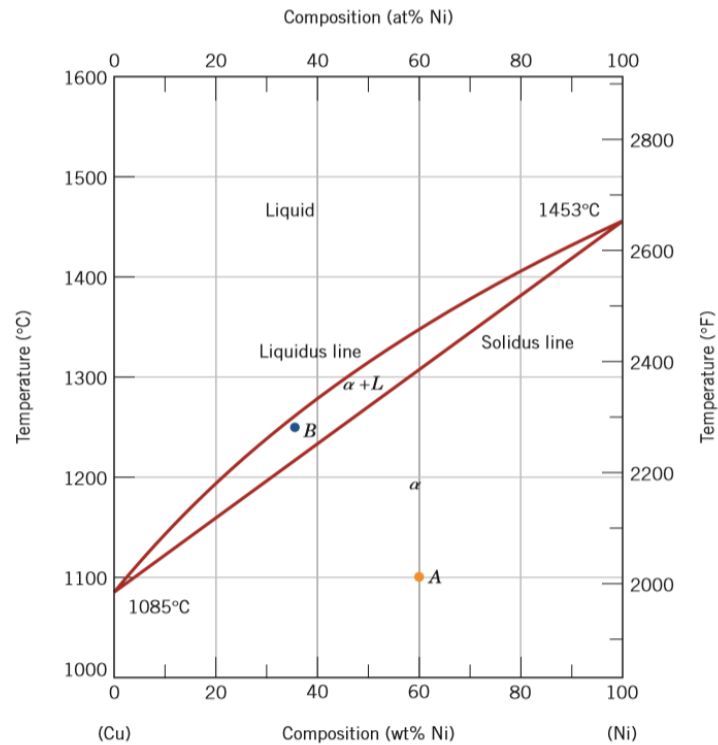
$$W_e = \frac{C_0 - C_\alpha}{C_{\text{eutectic}} - C_\alpha} = \frac{25 - 8.0}{71.9 - 8.0} = 0.266$$

(c) And, finally, we are asked to compute the mass fraction of eutectic α , $W_{e\alpha}$. This quantity is simply the difference between the mass fractions of total α and primary α as

$$W_{e\alpha} = W_\alpha - W_{\alpha'} = 0.796 - 0.734 = 0.062$$

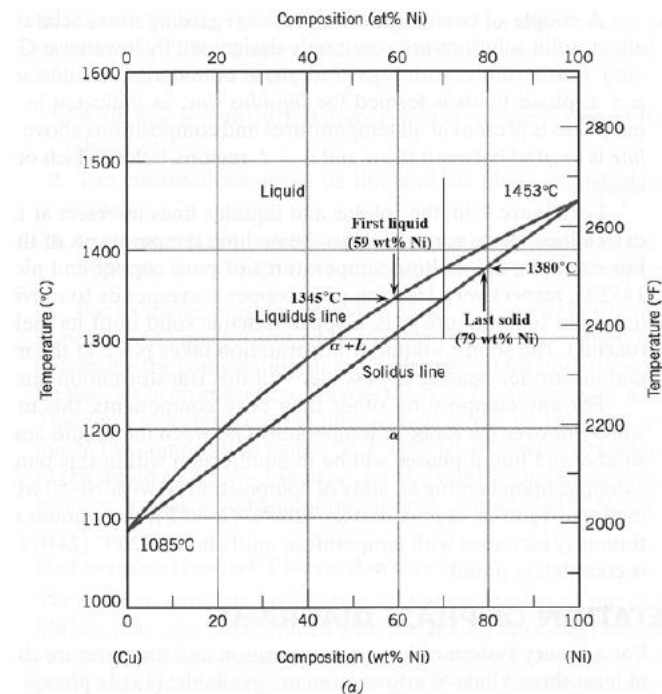
4. A copper-nickel alloy of composition 70 wt% Ni-30 wt% Cu is slowly heated from a temperature of 1300°C (2370°F).

- At what temperature does the first liquid phase form?
- What is the composition of this liquid phase?
- At what temperature does complete melting of the alloy occur?
- What is the composition of the last solid remaining prior to complete melting?



Solution

Shown below is the Cu-Ni phase diagram (Figure 9.3a) and a vertical line constructed at a composition of 70 wt% Ni-30 wt% Cu.



- Upon heating from 1300°C, the first liquid phase forms at the temperature at which this vertical line intersects the α -($\alpha + L$) phase boundary--i.e., about 1345°C.
- The composition of this liquid phase corresponds to the intersection with the ($\alpha + L$)-L phase boundary, of a tie line constructed across the $\alpha + L$ phase region at 1345°C--i.e., 59 wt% Ni;
- Complete melting of the alloy occurs at the intersection of this same vertical line at 70 wt% Ni with the ($\alpha + L$)-L phase boundary--i.e., about 1380°C;

(d) The composition of the last solid remaining prior to complete melting corresponds to the intersection with α -($\alpha + L$) phase boundary, of the tie line constructed across the $\alpha + L$ phase region at 1380°C--i.e., about 79 wt% Ni.

5. A 90 wt% Ag-10 wt% Cu alloy is heated to a temperature within the β + liquid phase region. If the composition of the liquid phase is 85 wt% Ag, determine:

- The temperature of the alloy
- The composition of the β phase
- The mass fractions of both phases

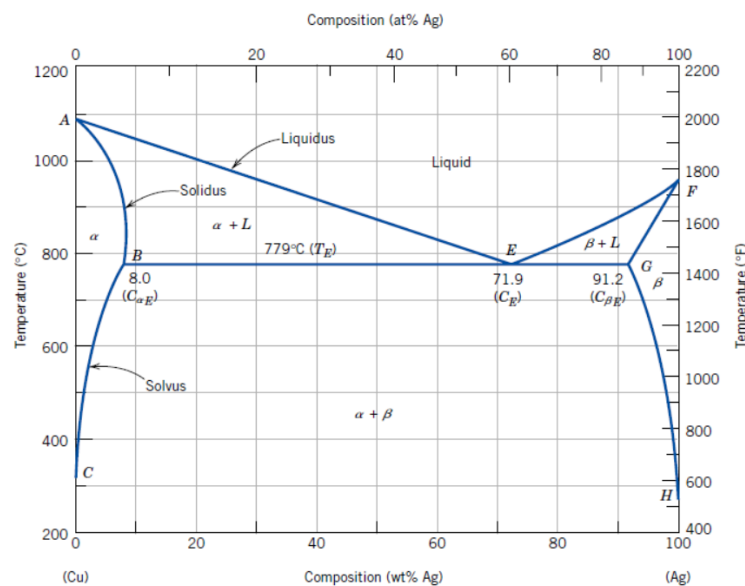


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Solution

(a) In order to determine the temperature of a 90 wt% Ag-10 wt% Cu alloy for which β and liquid phases are present with the liquid phase of composition 85 wt% Ag, we need to construct a tie line across the $\beta + L$ phase region of Figure 9.7 that intersects the liquidus line at 85 wt% Ag; this is possible at about 850°C.

(b) The composition of the β phase at this temperature is determined from the intersection of this same tie line with solidus line, which corresponds to about 95 wt% Ag.

(c) The mass fractions of the two phases are determined using the lever rule, Equations 9.1 and 9.2 with $C_0 = 90$ wt% Ag, $C_L = 85$ wt% Ag, and $C_\beta = 95$ wt% Ag, as

$$W_\beta = \frac{C_0 - C_L}{C_\beta - C_L} = \frac{90 - 85}{95 - 85} = 0.50$$

$$W_L = \frac{C_\beta - C_0}{C_\beta - C_L} = \frac{95 - 90}{95 - 85} = 0.50$$